

# Relation Algebra

## §14. Thinning Algorithms

### §14.1. Thinning

(14.1a)

For  $Q : A \rightarrow A$ ,  $\text{thin}(Q) \triangleq (\exists/\exists) \cap (\epsilon \setminus (Q^\circ \epsilon)) : EA \rightarrow EA$  (8.1).

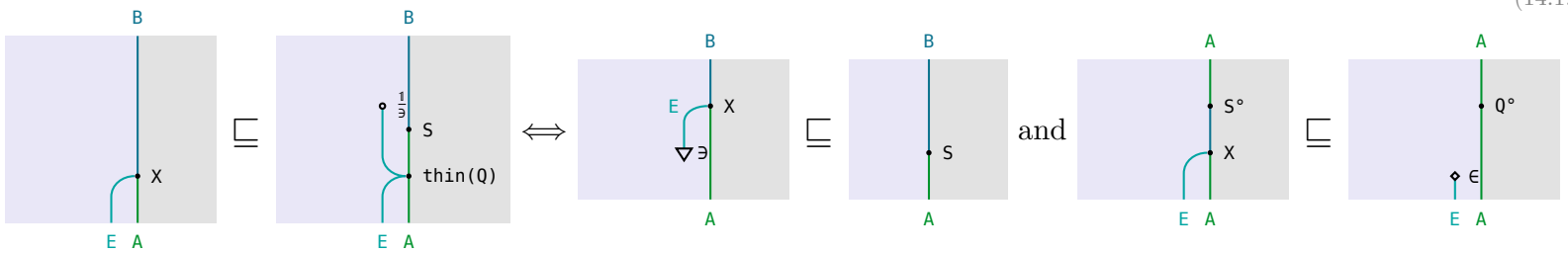
$ys \text{ thin}(Q) xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$  the same  $^\circ$  as  $\triangleq \text{est}$ :  $\epsilon X$  runs  $a \rightarrow ys \rightarrow xs$ , member of  $ys$  first, and  $Q^\circ \epsilon$  runs  $a \rightarrow b \rightarrow xs$ , so  $(a, b) \in Q^\circ$  reads  $b \ Q \ a$ ; at  $Q \triangleq \leq$  every  $a \in ys$  keeps some  $b \leq a$  in  $xs$ , the end  $\text{est}(\leq)$  picks

(14.1b)

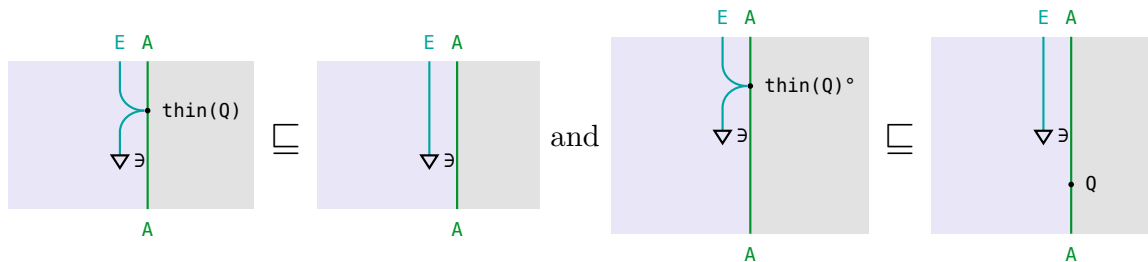
| the law                                                                                                                                                                                                                                                                        | what it says                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| $X \subseteq \frac{S}{\exists} \text{thin}(Q) \iff X \exists \in S$ and $S^\circ X \subseteq Q^\circ \epsilon$                                                                                                                                                                 | everything kept is an S-value, and every S-value has a Q-lower bound among the kept ones                             |
| $\text{thin}(Q) \exists \in \exists$                                                                                                                                                                                                                                           | everything a thinning keeps was in the set                                                                           |
| $\text{thin}(Q)^\circ \exists \in \exists Q$                                                                                                                                                                                                                                   | every element of the set has a Q-lower bound among the kept ones                                                     |
| $Q \subseteq R \implies \text{thin}(Q) \subseteq \text{thin}(R)$                                                                                                                                                                                                               | the fewer pairs Q relates, the fewer subsets count as thinnings                                                      |
| $1 \subseteq Q \implies 1 \subseteq \text{thin}(Q)$<br>$Q \subseteq Q \implies \text{thin}(Q) \text{thin}(Q) \subseteq \text{thin}(Q)$                                                                                                                                         | keeping everything is always a legal thinning                                                                        |
| $Q \subseteq R \wedge 1 \subseteq Q \wedge R R \subseteq R \implies \text{est}(R) = \text{thin}(Q) \text{est}(R)$ $Q \subseteq R, 1 \subseteq Q, R R \subseteq R$ — (14.1.2a); weaker than the book's “both preorders”: Q transitive is never used and $1 \subseteq R$ follows | <b>thin-introduction</b> : thinning first cannot lose an R-minimum                                                   |
| $\text{thin}(Q) \supseteq \text{est}(Q) \frac{1}{\exists}$ (8.2) — (14.1.2d)                                                                                                                                                                                                   | <b>thin-elimination</b> : keeping one element is a thinning, but its domain is the sets <b>est</b> (Q) is defined on |
| $\frac{S}{\exists} \text{thin}(Q) \supseteq \frac{S}{\exists} \text{est}(R) \frac{1}{\exists}$<br>(8.3), $R \cap (S^\circ S) \subseteq Q$ — thin-elimination variant                                                                                                           | the usable variant: R need only refine Q between values S gives one argument                                         |
| $P(\text{thin}(Q)) \text{union} \subseteq \text{union thin}(Q)$ (8.4)                                                                                                                                                                                                          | thinning each member set is a thinning of the union                                                                  |

#### §14.1.1. $X \subseteq (S^\circ \exists) \text{thin}(Q) \iff X \exists \in S$ and $S^\circ X \subseteq Q^\circ \epsilon$

(14.1.1a)



(14.1.1b)



(14.1.1a) at  $X \triangleq \text{thin}(Q)$ ,  $S \triangleq \exists$ : the left side is  $\text{thin}(Q) \subseteq \text{thin}(Q)$ , since  $\exists \exists = \wedge(\exists) = 1$

#### §14.1.2. $\text{est}(R) = \text{thin}(Q) \text{est}(R)$ given $Q \subseteq R, 1 \subseteq Q, R R \subseteq R$

(14.1.2a)

**$\text{est}(R) \sqsubseteq \text{thin}(Q) \text{ est}(R)$**

the  $\sqsubseteq$  half: keeping everything is a thinning —  $1 \sqsubseteq Q$

$EA \text{—} \boxed{\text{est}(R)} \text{—} A$   
 $\sqsubseteq$   
 $EA \text{—} \boxed{\text{thin}(Q)} \text{—} \boxed{\text{est}(R)} \text{—} A$   
 $\sqsubseteq$   
 $1 \sqsubseteq \text{thin}(Q) \text{ — (14.1b), } Q \text{ reflexive}$

(14.1.2b)

**$\text{thin}(Q) \text{ est}(R) \sqsubseteq \exists$**

the  $\sqsubseteq$  half, first condition of the UP of  $\text{est}$  at  $X \triangle \text{thin}(Q) \text{ est}(R)$  — est-up

$EA \text{—} \boxed{\text{thin}(Q)} \text{—} \boxed{\text{est}(R)} \text{—} A$   
 $\sqsubseteq$   
 $EA \text{—} \boxed{\text{thin}(Q)} \text{—} \boxed{\exists} \text{—} A$   
 $\sqsubseteq$   
 $\text{est}(R) \sqsubseteq \exists \text{ — est-laws first row at } X \triangle \text{est}(R)$   
 $\sqsubseteq$   
 $EA \text{—} \boxed{\exists} \text{—} A$   
 $\sqsubseteq$   
 $\text{thin}(Q) \exists \sqsubseteq \exists \text{ — (14.1.1a) at } S \triangle \exists, X \triangle \text{thin}(Q): \exists \exists = 1, \text{ so}$   
 $\text{the left side is } \text{thin}(Q) \sqsubseteq \text{thin}(Q)$

(14.1.2c)

**$\epsilon \text{ thin}(Q) \text{ est}(R) \sqsubseteq R^\circ$**

the  $\sqsubseteq$  half, second condition —  $Q \sqsubseteq R$ ,  $R$  transitive

$A \text{—} \boxed{\epsilon} \text{—} \boxed{\text{thin}(Q)} \text{—} EA \text{—} \boxed{\text{est}(R)} \text{—} A$   
 $\sqsubseteq$   
 $A \text{—} \boxed{Q} \text{—} \boxed{\epsilon} \text{—} EA \text{—} \boxed{\text{est}(R)} \text{—} A$   
 $\sqsubseteq$   
 $A \text{—} \boxed{Q} \text{—} \boxed{R} \text{—} A$   
 $\sqsubseteq$   
 $A \text{—} \boxed{R} \text{—} \boxed{R} \text{—} A$   
 $\sqsubseteq$   
 $A \text{—} \boxed{R} \text{—} A$

$\epsilon \text{ thin}(Q) \sqsubseteq Q^\circ \epsilon \text{ — (14.1.1a) at } S \triangle \exists, X \triangle \text{thin}(Q), \text{ second half}$   
 $\epsilon \text{ est}(R) \sqsubseteq R^\circ \text{ — est-laws first row at } X \triangle \text{est}(R), \text{ second half, conversed}$   
 $Q \sqsubseteq R, \text{ conversed}$   
 $R \text{ transitive, conversed}$

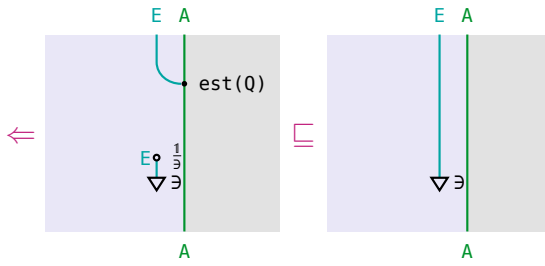
$$\text{est}(Q) \stackrel{1}{\exists} \text{thin}(Q)$$

(14.1.2d)

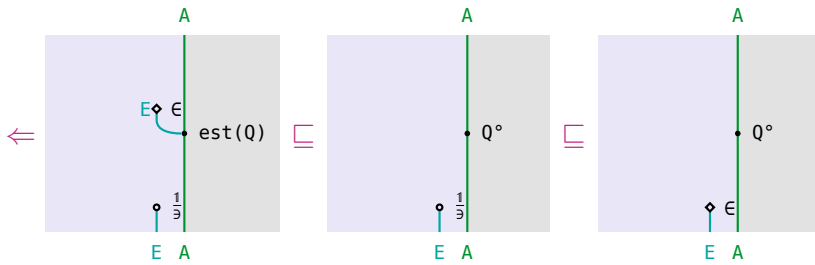
 the singleton holding a  $Q$ -least member of a set is a thinning of that set

$$\text{est}(Q) \stackrel{1}{\exists} \exists$$

every member of the singleton is a member of the set


 the  $\exists/\exists$  half of  $\trianglelefteq$  thin at  $X \trianglelefteq \text{est}(Q) \stackrel{1}{\exists}$ 
 $\stackrel{1}{\exists} \exists = 1$  — pow-laws — then  $\text{est}(Q) \sqsubseteq \exists$  — est-laws

$$\epsilon \text{est}(Q) \stackrel{1}{\exists} \sqsubseteq Q^\circ \epsilon$$

 the singleton's member stands in  $Q$  to every member of the set

 the  $\epsilon \backslash (Q^\circ \epsilon)$  half of  $\trianglelefteq$  thin,  $\neg \sqsubseteq Q^\circ \epsilon$ 
 $\epsilon \text{est}(Q) \sqsubseteq Q^\circ$  — est-up at  $X \trianglelefteq \text{est}(Q)$ , conversed

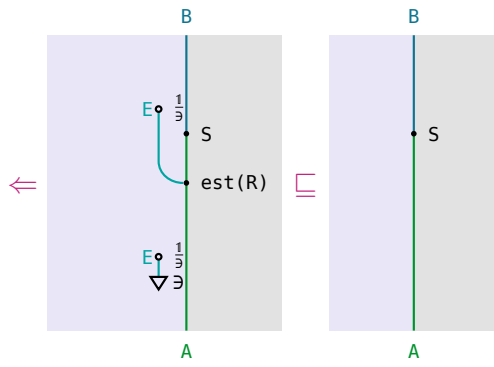
 $\stackrel{1}{\exists} \sqsubseteq \epsilon$ , since  $\stackrel{1}{\exists} \exists = 1$  with  $\stackrel{1}{\exists}$  a map — pow-laws

$$\text{RnS}^\circ \text{S} \sqsubseteq \text{Q} \Rightarrow \frac{\text{S}}{\exists} \text{est}(\text{R}) \frac{1}{\exists} \sqsubseteq \frac{\text{S}}{\exists} \text{thin}(\text{Q})$$

(14.1.2e)

 given  $\text{Rn}(\text{S}^\circ \text{S}) \sqsubseteq \text{Q}$ ,  $\text{Q}$  a preorder

$$\frac{\text{S}}{\exists} \text{est}(\text{R}) \frac{1}{\exists} \sqsubseteq \text{S}$$

 every member of the singleton is a value  $\text{S}$  returns


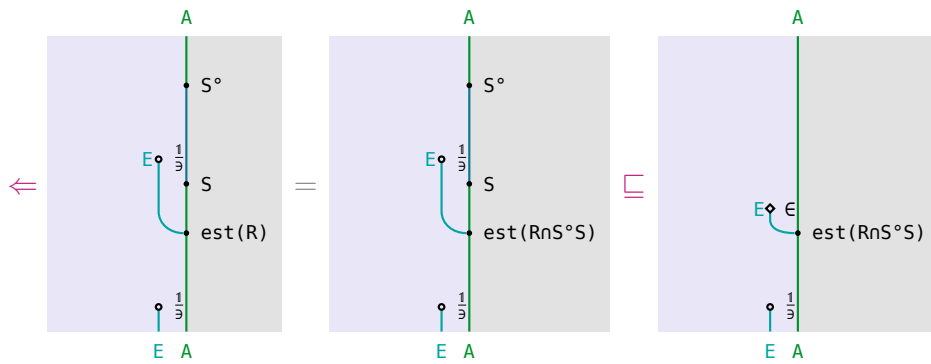
$$\leftarrow B \xrightarrow{\frac{1}{\exists}} EB \xrightarrow{E(S)} EA \xrightarrow{\text{est}(\text{R})} A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{\exists} A$$

 the first of (14.1b)' two conditions at  $X \triangleq \frac{\text{S}}{\exists} \text{est}(\text{R}) \frac{1}{\exists}$ 

$$\sqsubseteq B \xrightarrow{\text{S}} A$$

 $\frac{1}{\exists} \sqsubseteq 1$  — pow-laws — then  $\frac{\text{S}}{\exists} \text{est}(\text{R}) = \text{Sn}(\text{S}^\circ \setminus \text{R}^\circ) \sqsubseteq \text{S}$  — est-laws

$$\text{RnS}^\circ \text{S} \sqsubseteq \text{Q} \Rightarrow \text{S}^\circ \frac{\text{S}}{\exists} \text{est}(\text{R}) \frac{1}{\exists} \sqsubseteq \text{Q}^\circ \sqsubseteq \text{E}$$

 the singleton's member stands in  $\text{Q}$  to every value  $\text{S}$  returns at the same argument


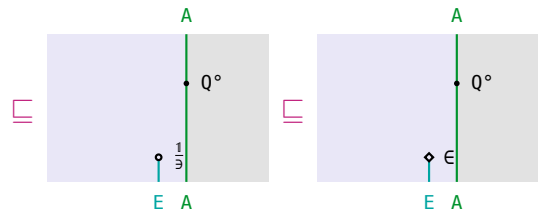
$$\leftarrow A \xrightarrow{\text{S}} B \xrightarrow{\frac{1}{\exists}} EB \xrightarrow{E(S)} EA \xrightarrow{\text{est}(\text{R})} A \xrightarrow{\frac{1}{\exists}} EA$$

 the second condition,  $\neg \text{Q}^\circ \sqsubseteq \text{E}$ 

$$= A \xrightarrow{\text{S}} B \xrightarrow{\frac{1}{\exists}} EB \xrightarrow{E(S)} EA \xrightarrow{\text{est}(\text{RnS}^\circ \text{S})} A \xrightarrow{\frac{1}{\exists}} EA$$

 $\frac{\text{S}}{\exists} \text{est}(\text{R}) = \frac{\text{S}}{\exists} \text{est}(\text{RnS}^\circ \text{S})$  — est-laws

$$\sqsubseteq A \xrightarrow{\text{E}} EA \xrightarrow{\text{est}(\text{RnS}^\circ \text{S})} A \xrightarrow{\frac{1}{\exists}} EA$$

 $\text{S}^\circ \frac{\text{S}}{\exists} \sqsubseteq \text{E}$ , since  $\frac{\text{S}}{\exists} \sqsubseteq \text{S}$  with  $\frac{\text{S}}{\exists}$  a map — pow-laws


$$\sqsubseteq A \xrightarrow{\text{Q}} \frac{1}{\exists} \xrightarrow{\text{E}} EA$$

 $\text{E} \text{est}(\text{RnS}^\circ \text{S}) \sqsubseteq (\text{RnS}^\circ \text{S})^\circ$  — est-up at  $X \triangleq \text{est}(\text{RnS}^\circ \text{S})$ , conversed — then  $\text{Rn}(\text{S}^\circ \text{S}) \sqsubseteq \text{Q}$ 

$$\sqsubseteq A \xrightarrow{\text{Q}} \text{E} \xrightarrow{\text{E}} EA$$

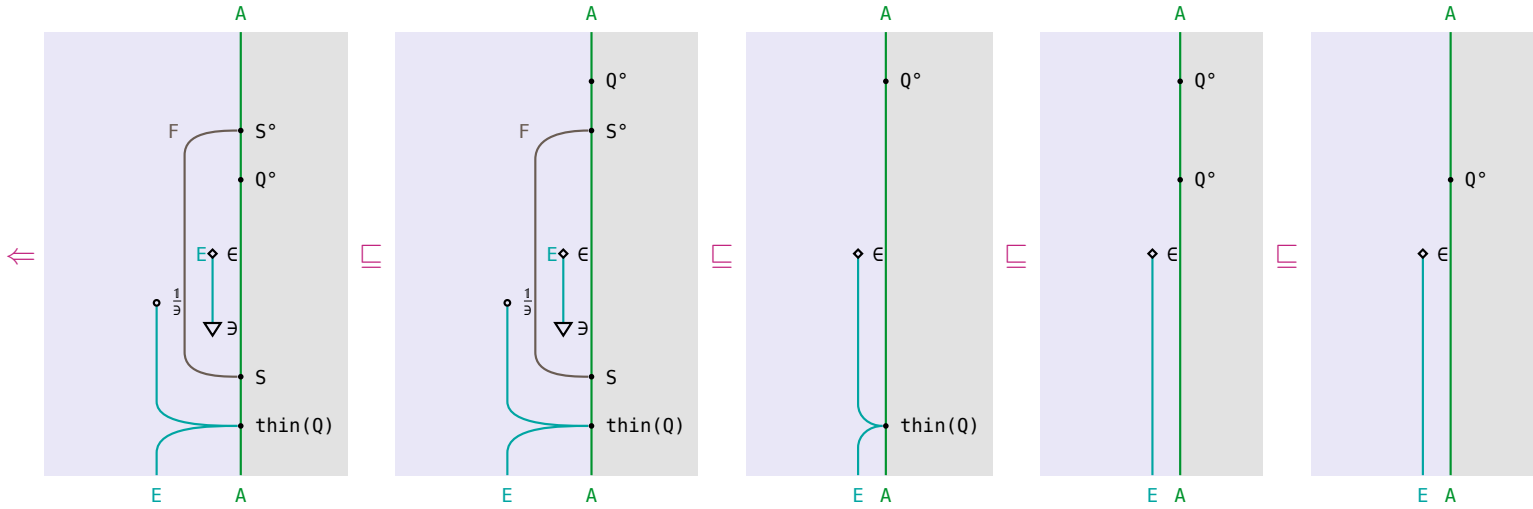
 $\frac{1}{\exists} \sqsubseteq \text{E}$ , since  $\frac{1}{\exists} \sqsubseteq 1$  with  $\frac{1}{\exists}$  a map — pow-laws

$$\text{PreservesRecip}(F) \wedge Q \subseteq Q \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \subseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle \quad (14.1.2f)$$

thinning at every step of the reduce is a thinning of the whole candidate set —  $S$  monotonic on  $Q$ ,  $Q$  a preorder

$$\langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \subseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle \quad \text{and} \quad \langle \frac{(S)}{\exists} \rangle \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \subseteq Q^\circ \epsilon$$

(14.1b) at  $X \triangleq \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle$ ,  $\langle \frac{(S)}{\exists} \rangle$  for its  $S$



$$A \rightarrow [S] \rightarrow FA \rightarrow [F(Q^\circ \epsilon)] \rightarrow FEA \rightarrow [\frac{1}{\exists}] \rightarrow EFEA \rightarrow [E(F(\exists)S)] \rightarrow \text{thin}(Q) \rightarrow EA$$

the first by fusion; hylo-least at the bound  $Q^\circ \epsilon$  reduces the second to  $\neg \epsilon Q^\circ \epsilon$

$$A \rightarrow [Q] \rightarrow [S] \rightarrow FA \rightarrow [F(\epsilon)] \rightarrow FEA \rightarrow [\frac{1}{\exists}] \rightarrow EFEA \rightarrow [E(F(\exists)S)] \rightarrow \text{thin}(Q) \rightarrow EA$$

$S^\circ F(Q^\circ) \subseteq Q^\circ S^\circ$  — mon-str at  $S$ , conversed;  $F(R)^\circ = F(R^\circ)$  — relator-laws

$$A \rightarrow [Q] \rightarrow [\epsilon] \rightarrow \text{thin}(Q) \rightarrow EA$$

$S^\circ F(\epsilon) \subseteq \frac{F(\exists)S}{\exists} \subseteq \epsilon$ , since  $\frac{F(\exists)S}{\exists} \subseteq F(\exists)S$  with  $\frac{F(\exists)S}{\exists}$  a map — pow-laws

$$A \rightarrow [Q] \rightarrow [Q] \rightarrow [\epsilon] \rightarrow EA$$

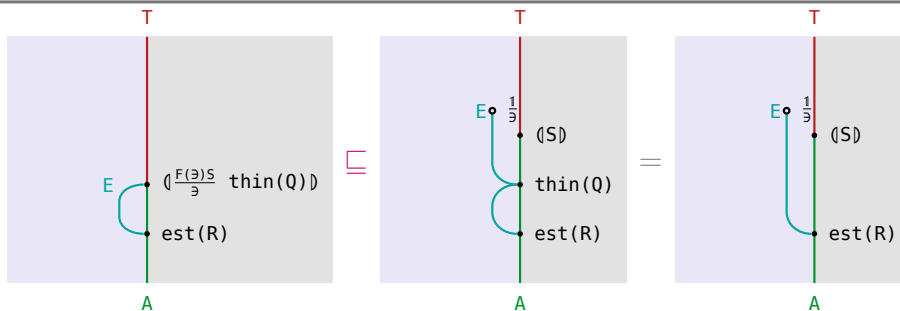
$\epsilon \text{thin}(Q) \subseteq Q^\circ \epsilon$ , the  $\epsilon \setminus (Q^\circ \epsilon)$  half of  $\triangle \text{thin}$  — adj-all

$$A \rightarrow [Q] \rightarrow [\epsilon] \rightarrow EA$$

$Q^\circ Q^\circ = Q^\circ$ ,  $Q$  a preorder

$$\text{PreservesRecip}(F) \wedge Q \subseteq R \wedge 1 \subseteq Q \wedge Q \subseteq Q \wedge R \subseteq R \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \text{est}(R) \subseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle \text{est}(R) \quad (14.1.2g)$$

the thinning fold refines the optimisation problem itself —  $S$  monotonic on  $Q$ ,  $Q \subseteq R$ , both preorders



$$T \rightarrow [FEA] \rightarrow [\frac{1}{\exists}] \rightarrow EFEA \rightarrow [E(F(\exists)S)] \rightarrow \text{thin}(Q) \rightarrow EA \rightarrow \text{est}(R) \rightarrow A$$

$$T \rightarrow [\frac{1}{\exists}] \rightarrow E(T) \rightarrow [E(\langle \frac{(S)}{\exists} \rangle)] \rightarrow \text{thin}(Q) \rightarrow EA \rightarrow \text{est}(R) \rightarrow A$$

thinning theorem

$$T \rightarrow [\frac{1}{\exists}] \rightarrow E(T) \rightarrow [E(\langle \frac{(S)}{\exists} \rangle)] \rightarrow EA \rightarrow \text{est}(R) \rightarrow A$$

$$\text{est}(R) = \text{thin}(Q) \text{est}(R) \quad (14.1b), Q \subseteq R$$

## §14.2. Paths in a layered network

(14.2a)

$F(A, X) = A + A \times X$ ,  $L = \text{list}^+$  with initial algebra  $\alpha \triangleq [\text{wrap}, \text{cons}] : F(A, LA) \rightarrow LA$ .

$\text{wrapz} \triangleq \langle \text{wrap}, \text{zero} \rangle$ ,  $\text{consw}(a, (xs, n)) = (\text{cons}(a, xs), \text{wt}(a, \text{head}(xs)) + n)$ .

$\text{cost} \triangleq \langle [\text{wrapz}, \text{consw}] \rangle \pi_2$ ,  $\langle [\text{wrapz}, \text{consw}] \rangle = \langle 1, \text{cost} \rangle$ ,  $R \triangleq \text{cost} \leq \text{cost}^\circ$ .

$Q \triangleq R \cap (\text{head} \text{head}^\circ)$ ,  $S \triangleq F(1, \exists) \alpha$ ,  $\frac{F(\exists, 1)}{\exists} = 1 + \text{cpl}$ ,  $\frac{F(1, \exists)}{\exists} = 1 + \text{cpr}$ ,  $\text{step} \triangleq \text{cpr } P(\text{cons}) \text{est}(R)$ .

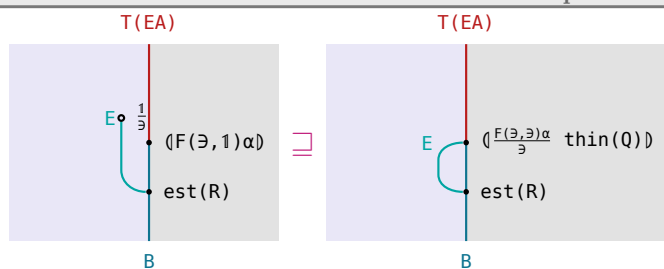
(14.2b)

| the law                                                                         | what it says                                                                                                                                                        |
|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $F(\exists, Q) \alpha \sqsubseteq F(\exists, 1) \alpha Q$                       | $F(\exists, 1) \alpha$ is monotonic on $Q$ ; on $R$ it is not, since the next edge can cost arbitrarily much                                                        |
| $S \text{ head} \sqsubseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$ | $S \text{ head}$ is simple, which gives $Rn(S \circ S) \sqsubseteq Q$ : between two paths $S$ builds from one argument, equal cost and equal head already means $Q$ |

$$\text{PreservesRecip}(F) \wedge Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R \circ R \circ \sqsubseteq R \circ \wedge \text{MonoAlg}(F(\exists, 1) \alpha, Q) \wedge Rn(F(1, \exists) \alpha) \circ F(1, \exists) \alpha \sqsubseteq Q \quad (14.2c)$$

$$\Rightarrow \langle \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R)) \rangle \text{est}(R) \sqsubseteq \langle \frac{F(\exists, 1) \alpha}{\exists} \rangle \text{est}(R)$$

a least-cost path in a layered network, as a fold over the layers



$$\begin{aligned} & T(EA) \rightarrow \left[ \frac{1}{\exists} \right] \rightarrow E(T(EA)) \rightarrow E(F(\exists, 1) \alpha) \rightarrow EB \rightarrow \text{est}(R) \rightarrow B \\ & \sqsubseteq T(EA) \rightarrow \left[ \text{FEB} \right] \left[ \frac{1}{\exists} \right] \rightarrow EFEB \rightarrow E(F(\exists, \exists) \alpha) \rightarrow \text{thin}(Q) \rightarrow EB \rightarrow \text{est}(R) \rightarrow B \end{aligned} \quad = \frac{L(\exists)}{\exists} \text{est}(R)$$

 (14.1.2g), at  $F(\exists, 1) \alpha$  monotonic on  $Q$  — (14.2b).

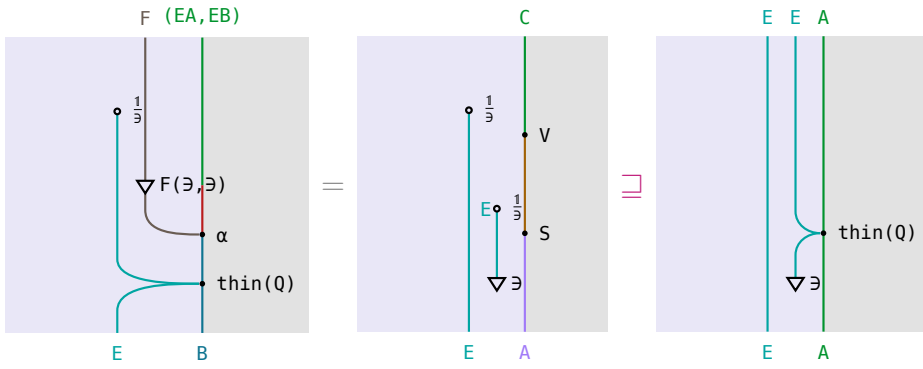
 $\sqsubseteq$  (14.2d) under  $\langle \rangle$  monotonic: the whole chain runs inside the reduce, and the  $\text{est}(R)$  behind it never moves.

$$Rn(F(1, \exists)\alpha) \circ F(1, \exists)\alpha \sqsubseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists} \text{est}(R)) \sqsubseteq \frac{F(\exists, \exists)\alpha}{\exists} \text{thin}(Q) \quad (14.2d)$$

thinning the algebra of a layered network costs no more than taking the program's two cases

$$\frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) P(\text{thin}(Q)) \text{union} \sqsubseteq \frac{F(\exists, \exists)\alpha}{\exists} \text{thin}(Q)$$

thinning each set the algebra's parts build, then taking the union, is below thinning the one set the whole algebra builds



$$F(EA)(EB) \xrightarrow{\frac{1}{\exists}} E(F(EA)(EB)) \xrightarrow{E(F(\exists, \exists)\alpha)} \text{thin}(Q) \xrightarrow{} EB$$

the algebra of (14.2c) row 2

$$= C \xrightarrow{\frac{1}{\exists}} EC \xrightarrow{E(V)} EW \xrightarrow{P(\frac{S}{\exists})} EEA \xrightarrow{\text{union}} EA$$

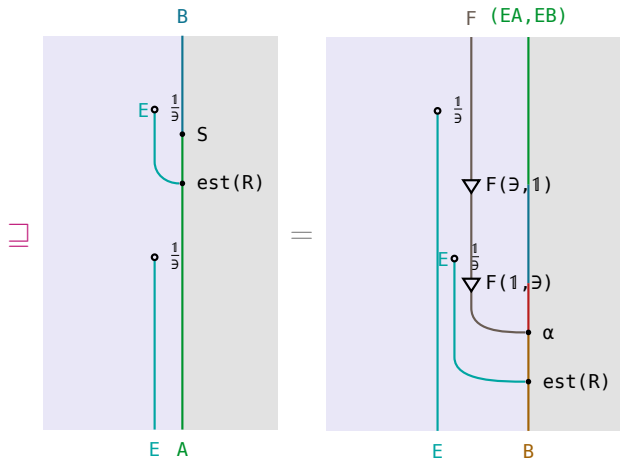
$$F(\exists, \exists) = F(\exists, 1)F(1, \exists); \quad \frac{F(\exists, 1)F(1, \exists)\alpha}{\exists} = \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \text{union}.$$

$$\sqsubseteq EEA \xrightarrow{P(\text{thin}(Q))} \text{union} \xrightarrow{} EA$$

$$\text{union thin}(Q) \sqsupseteq P(\text{thin}(Q)) \text{union} \text{ --- (14.1b).}$$

$$Rn(F(1, \exists)\alpha) \circ F(1, \exists)\alpha \sqsubseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists} \text{est}(R)) \sqsubseteq \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) P(\text{thin}(Q)) \text{union}$$

choosing an R-least member of each of those sets is below thinning each and taking the union



$$\sqsubseteq B \xrightarrow{\frac{1}{\exists}} EB \xrightarrow{E(S)} EA \xrightarrow{\text{est}(R)} A \xrightarrow{\frac{1}{\exists}} EA$$

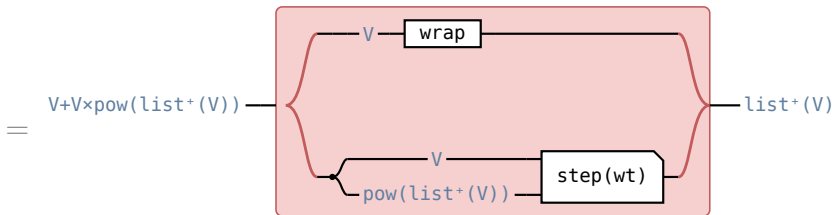
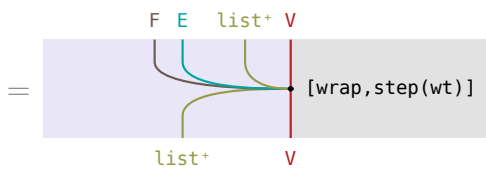
$$\frac{S}{\exists} \text{thin}(Q) \sqsupseteq \frac{S}{\exists} \text{est}(R) \xrightarrow{\frac{1}{\exists}} \text{ --- (14.1b) at } S \triangleq F(1, \exists)\alpha, Rn(S \circ S) \sqsubseteq Q \text{ --- (14.2b).}$$

$$= F(EA)(EB) \xrightarrow{\frac{1}{\exists}} E(F(EA)(EB)) \xrightarrow{E(F(\exists, 1))} E(FA)(EB) \xrightarrow{P(\frac{F(1, \exists)\alpha}{\exists} \text{est}(R))} EB$$

$$P(\frac{1}{\exists}) \text{union} = 1.$$

$$\text{cp}(F, \text{list}^+(V)) P(\alpha) \text{est}(\text{path}(\text{wt})) = [\text{wrap}, \text{step}(\text{wt})]$$

at the layered network, choosing an R-least path from each case is the program's two cases  $[\text{wrap}, \text{step}]$



$$\alpha \text{ a map, so } \frac{F(1, \exists)\alpha}{\exists} = \frac{F(1, \exists)}{\exists} P(\alpha); \quad \frac{F(1, \exists)}{\exists} P(\alpha) \text{est}(R) = [\text{wrap}, \text{step}] \triangleq \text{cost}.$$

$$= \frac{F(\exists, 1)}{\exists} = 1 + \text{cpl} \triangleq \text{cost}$$

## §14.3. Implementing thin

(14.3a)

$\text{setify} : [A] \rightarrow EA$ ,  $\text{cup} : EA \times EA \rightarrow EA$ ,  $\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$ ,  $\text{listcp} : F(L) \rightarrow LF$ ,  $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P$  for  $P$  a connected preorder.

$\text{thinlist}(Q)$  is any  $\text{thinlist}(Q) \sqsubseteq \text{subseq}$  with  $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q)$ ; one is  $\langle [\text{nil}, \text{bump } Q] \rangle$ ,  $\text{bump } Q (a, []) = [a]$ ,  $\text{bump } Q (a, [b] \# xs) = (b \text{ } Q \text{ } a \rightarrow [a] \# xs, a \text{ } Q \text{ } b \rightarrow [b] \# xs, [a] \# [b] \# xs)$ .

**Binary thinning** data:  $S = (f_1 p_1) \cup (f_2 p_2)$  with  $p_1, p_2$  coreflexive;  $Q$  a preorder with  $Q \sqsubseteq R$  and both  $f_1 p_1, f_2 p_2$  monotonic on  $Q$ ;  $P$  a connected preorder with both  $f_1, f_2$  monotonic on  $P$ ;  $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$ .

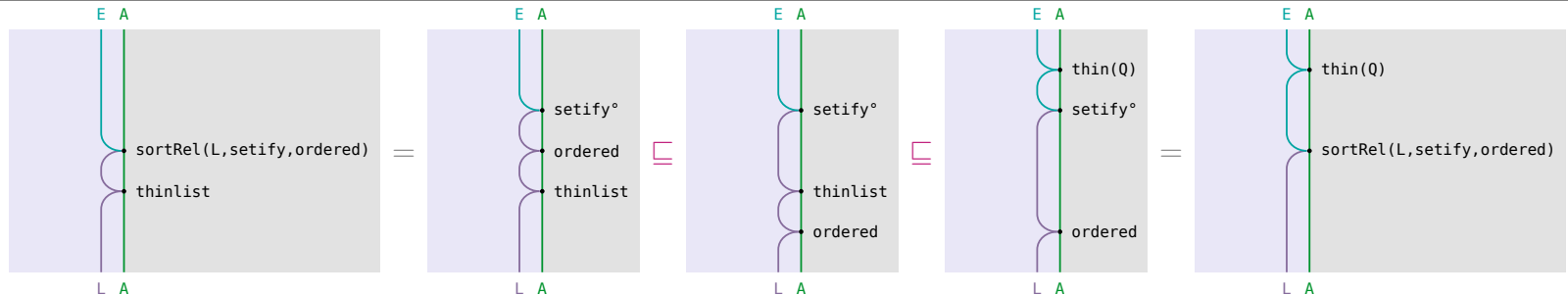
(14.3b)

| the law                                                                                                         | what it says                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| $\text{thinlist}(Q) \text{ xs} = [\text{minlist}(Q) \text{ xs}]$<br>(8.5), $Q$ connected, $\text{xs}$ non-empty | what thinning should come to when it can: one element                                                              |
| $\text{sort}(P) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(P)$ (8.6) — (14.3c)                 | thinning a sorted list is a thinning of the set — this is what $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ buys |
| $\text{sort}(P) \text{ minlist}(Q) \sqsubseteq \text{est}(Q)$ (8.7)                                             | a minimum of the sorted list is a minimum of the set                                                               |
| $\text{sort}(f P f^\circ) \text{ list}(f) \sqsubseteq P(f) \text{ sort}(P)$ (8.8)                               | shunt a function through a sort                                                                                    |
| $\text{sort}(P) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(P)$ (8.9), $p$ coreflexive                      | filtering a sorted list sorts the restricted set                                                                   |
| $(\text{sort}(P) \times \text{sort}(P)) \text{ merge}(P) \sqsubseteq \text{cup sort}(P)$ (8.10)                 | merging two sorted lists sorts their union                                                                         |
| $F(\text{sort}(P)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(FP)$<br>(8.11), $F$ linear              | $\text{listcp}$ is the list implementation of the cartesian product $\text{cp}(F)$                                 |

**Map(setify)  $\wedge$  Coreflexive(ordered)  $\wedge$  thinlist  $\sqsubseteq$  subseq  $\wedge$  ordered subseq  $\sqsubseteq$  subseq ordered  $\wedge$**  (14.3c)

**thinlist setify  $\sqsubseteq$  setify thin(Q)  $\Rightarrow$  sortRel(L, setify, ordered) thinlist  $\sqsubseteq$  thin(Q) sortRel(L, setify, ordered)**

a thinning of the sorted list lists a thinning of the set —  $P$  a connected preorder,  $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ .



$EA \xrightarrow{\text{setify}} \boxed{\text{ordered}} \xrightarrow{\text{thinlist}} LA$

$= EA \xrightarrow{\text{setify}} \boxed{\text{ordered}} \xrightarrow{\text{thinlist}} LA$

$EA \xrightarrow{\text{setify}} \boxed{\text{thinlist}} \xrightarrow{\text{ordered}} LA$

$\sqsubseteq$  ordered  $P$   $\text{thinlist}(Q) \sqsubseteq \text{thinlist}(Q)$  ordered  $P$ , since  $\text{thinlist}(Q) \sqsubseteq \text{subseq} \triangleq \text{thinlist}$  — and a subsequence of a  $P$ -ordered list is  $P$ -ordered

$EA \xrightarrow{\text{thin}(Q)} \boxed{\text{setify}} \xrightarrow{\text{ordered}} LA$

$\sqsubseteq$   $\triangleq$  thinlist's  $\text{thinlist}(Q)$   $\text{setify} \sqsubseteq \text{setify thin}(Q)$  after  $\text{setify}^\circ$ , at  $\text{setify}^\circ \text{ setify} \sqsubseteq 1$  for  $\text{setify}$  simple — dom-laws — then  $\cdot \text{setify} \cdot \text{setify}^\circ$  — triple-chains

$= EA \xrightarrow{\text{thin}(Q)} \boxed{\text{setify}} \xrightarrow{\text{ordered}} LA$

$\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \triangleq \text{thinlist}$

$\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \triangleq \text{thinlist}$



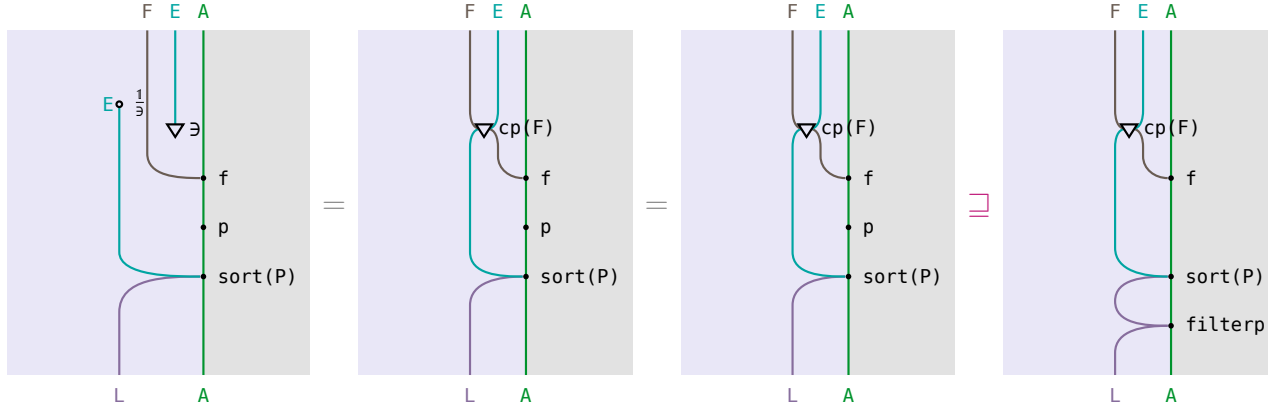
$$\text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f, P) \wedge \text{sortF}(fP f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(P) \wedge \quad (14.3d)$$

$$\text{sort}(P) \text{filterp} \sqsubseteq E(p) \text{sort}(P) \wedge F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}(F, A) \text{sortF}(F(P)) \Rightarrow F(\text{sort}(P)) \text{listcp} \text{listf} \text{filterp} \sqsubseteq \frac{F(\exists)fp}{\exists} \text{sort}(P)$$

one sorted list built from sorted arguments, instead of a set built and then sorted —  $f : FA \rightarrow A$  monotonic on  $P$ ,  $p$  coreflexive,  $F$  linear.

$$\text{Map}(f) \wedge \text{sort}(P) \text{filterp} \sqsubseteq E(p) \text{sort}(P) \Rightarrow \text{cp}(F, A) P(f) \text{sort}(P) \text{filterp} \sqsubseteq \frac{F(\exists)fp}{\exists} \text{sort}(P)$$

sorting the  $f$ -images of the product and filtering by  $p$  is below sorting the set  $F(\exists)fp$  builds

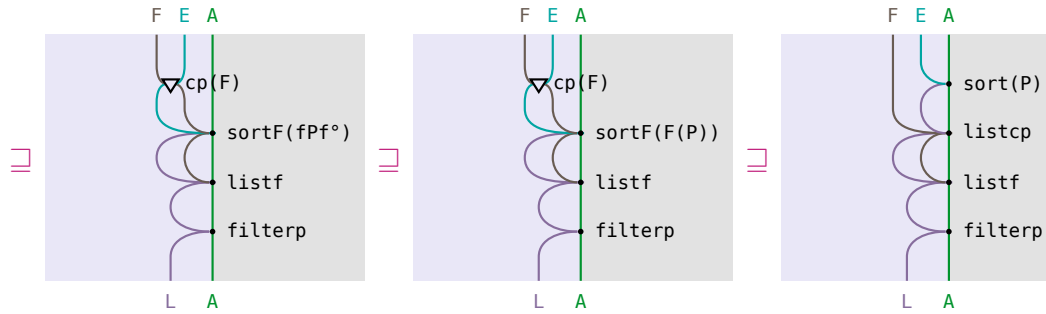


$$\begin{aligned} & \text{FEA} \xrightarrow{\frac{1}{\exists}} \text{EFEA} \xrightarrow{E(F(\exists)fp)} \text{EA} \xrightarrow{\text{sort}(P)} \text{LA} \\ = & \text{FEA} \xrightarrow{\text{cp}(F, A)} \text{EFA} \xrightarrow{E(fp)} \text{EA} \xrightarrow{\text{sort}(P)} \text{LA} \\ = & \text{FEA} \xrightarrow{\text{cp}(F, A)} \text{EFA} \xrightarrow{P(f)} \text{EA} \xrightarrow{\text{sort}(P)} \text{LA} \\ \sqsubseteq & \text{FEA} \xrightarrow{\text{cp}(F, A)} \text{EFA} \xrightarrow{P(f)} \text{EA} \xrightarrow{\text{sort}(P)} \text{filterp} \xrightarrow{} \text{LA} \end{aligned}$$

$$\begin{aligned} \frac{F(\exists)fp}{\exists} &= \frac{F(\exists)}{\exists} E(fp) \text{ — pow-laws; } \text{cp}(F) \triangleq \frac{F(\exists)}{\exists} \text{ — thinlist} \\ E(fp) &= E(f)E(p); E(f) = P(f) \text{ for } f \text{ a map — powrel-laws} \\ \text{sort}(P) \text{ filter}(p) &\sqsubseteq E(p) \text{ sort}(P) \text{ — (14.3b)} \end{aligned}$$

$$\text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f, P) \wedge \text{sortF}(fP f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(P) \wedge F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}(F, A) \text{sortF}(F(P)) \Rightarrow F(\text{sort}(P)) \text{listcp} \text{listf} \text{filterp} \sqsubseteq \text{cp}(F, A) P(f) \text{sort}(P) \text{filterp}$$

combining the arguments' sorted lists by  $\text{listcp}$ , mapping  $f$  and filtering is below sorting the  $f$ -images and filtering



$$\begin{aligned} \sqsubseteq & \text{FEA} \xrightarrow{\text{cp}(F, A)} \text{EFA} \xrightarrow{\text{sortF}(fP f^\circ)} \text{LFA} \xrightarrow{\text{listf}} \text{filterp} \xrightarrow{} \text{LA} \\ \sqsubseteq & \text{FEA} \xrightarrow{\text{cp}(F, A)} \text{EFA} \xrightarrow{\text{sortF}(F(P))} \text{LFA} \xrightarrow{\text{listf}} \text{filterp} \xrightarrow{} \text{LA} \\ \sqsubseteq & \text{FEA} \xrightarrow{F(\text{sort}(P))} \text{FLA} \xrightarrow{\text{listcp}} \text{LFA} \xrightarrow{\text{listf}} \text{filterp} \xrightarrow{} \text{LA} \end{aligned}$$

$$\text{sort}(fP f^\circ) \text{list}(f) \sqsubseteq P(f) \text{sort}(P) \text{ — (14.3b)}$$

$$FP \sqsubseteq fP f^\circ \text{ — mon-str at } f \text{ a map; } \text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \text{ grows with } P \text{ — thinlist}$$

$$F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sort}(FP) \text{ — (14.3b), } F \text{ linear}$$

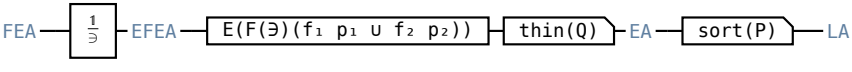
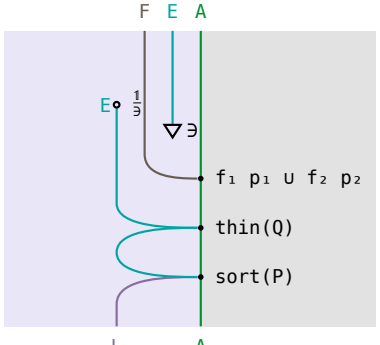
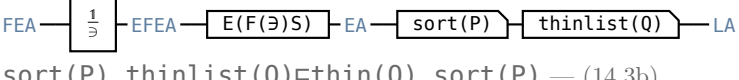
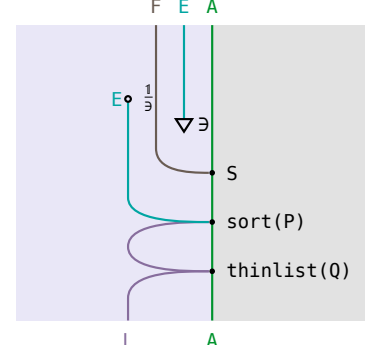
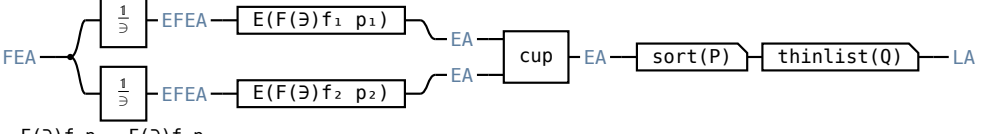
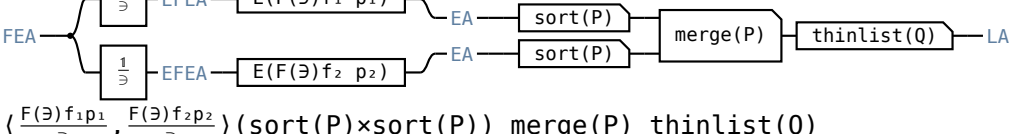
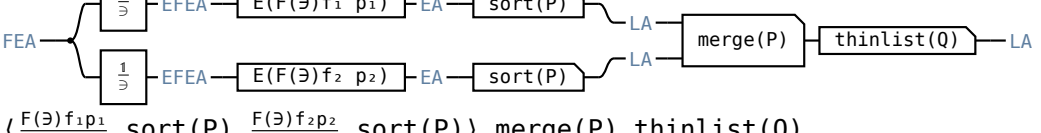
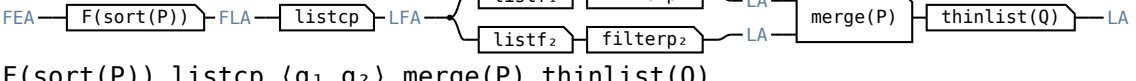
$$\begin{aligned}
& \text{PreservesRecip}(F) \wedge \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R^\circ R^\circ \sqsubseteq R^\circ \wedge \text{MonoAlg}(f_1, p_1, Q) \wedge \\
& \text{MonoAlg}(f_2, p_2, Q) \wedge (\forall X. \forall Y. X \sqsubseteq Y \implies \text{sort}F(X) \sqsubseteq \text{sort}F(Y)) \wedge \text{MonoAlg}(f_1, P) \wedge \text{MonoAlg}(f_2, P) \wedge \\
& \text{sort}F(f_1 \text{ P } f_1^\circ) \text{list}f_1 \sqsubseteq P(f_1) \text{sort}(P) \wedge \text{sort}F(f_2 \text{ P } f_2^\circ) \text{list}f_2 \sqsubseteq P(f_2) \text{sort}(P) \wedge \\
& \text{sort}(P) \text{filter}p_1 \sqsubseteq E(p_1) \text{sort}(P) \wedge \text{sort}(P) \text{filter}p_2 \sqsubseteq E(p_2) \text{sort}(P) \wedge \\
& F(\text{sort}(P)) \text{list}cp \sqsubseteq cp(F, A) \text{sort}F(F(P)) \wedge (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \sqsubseteq \text{cup} \text{sort}(P) \wedge \\
& \text{sort}(P) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(P) \wedge \text{sort}(P) \text{minlist}(R) \sqsubseteq \text{est}(R) \wedge g_1 = \text{list}f_1 \text{ filter}p_1 \wedge \\
& g_2 = \text{list}f_2 \text{ filter}p_2 \wedge S = f_1 \text{ p }_1 \cup f_2 \text{ p }_2 \implies (\text{list}cp(g_1, g_2) \text{merge}(P) \text{thinlist}(Q)) \text{minlist}(R) \sqsubseteq \frac{(S)}{3} \\
& \text{est}(R)
\end{aligned} \tag{14.3e}$$

a fold on sorted lists of partial solutions, thinned at every step — at  $\triangleq \text{thinlist}$ 's binary thinning data.

| circuit                                                                                                                      | Hinze–Marsden |
|------------------------------------------------------------------------------------------------------------------------------|---------------|
|                                                                                                                              |               |
| <p>(14.1.2g) at <math>f_1 p_1</math> and <math>f_2 p_2</math> monotonic on <math>Q \triangleq \text{thinlist}</math></p>     |               |
| <p><math>\text{sort}(P) \text{ minlist}(R) \sqsubseteq \text{est}(R)</math> — (14.3b) at its <math>Q \triangleq R</math></p> |               |
| <p>fusion at (14.3f)</p>                                                                                                     |               |

$$\begin{aligned}
& \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, P) \wedge \text{MonoAlg}(f_2, P) \wedge \quad (14.3f) \\
& \text{sortF}(f_1 \text{ Pf}_1^\circ) \text{listf}_1 \sqsubseteq \text{P}(f_1) \text{sort}(P) \wedge \text{sortF}(f_2 \text{ Pf}_2^\circ) \text{listf}_2 \sqsubseteq \text{P}(f_2) \text{sort}(P) \wedge \\
& \text{sort}(P) \text{filterp}_1 \sqsubseteq \text{E}(p_1) \text{sort}(P) \wedge \text{sort}(P) \text{filterp}_2 \sqsubseteq \text{E}(p_2) \text{sort}(P) \wedge \\
& \text{F}(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}(\text{F}, \text{A}) \text{sortF}(\text{F}(P)) \wedge (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \sqsubseteq \text{cup} \text{sort}(P) \wedge \\
& \text{sort}(P) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(P) \Rightarrow \text{F}(\text{sort}(P)) \text{listcp}(\text{listf}_1 \text{ filterp}_1, \text{listf}_2 \text{ filterp}_2) \\
& \text{merge}(P) \text{thinlist}(Q) \sqsubseteq \frac{\text{F}(\exists)(f_1 \text{ p}_1 \cup f_2 \text{ p}_2)}{\exists} \text{thin}(Q) \text{sort}(P)
\end{aligned}$$

sorting the candidate set is what turns the thinning algebra into an algebra on lists — the side condition of binary thinning theorem's last step.

| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Hinze–Marsden                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                       |   |
| $\sqsubseteq$ <br>$\text{sort}(P) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(P) \text{ — (14.3b)}$                                                                                                                                                                                                                                                                     |  |
| $=$ <br>$\langle \frac{\text{F}(\exists)f_1 p_1}{\exists}, \frac{\text{F}(\exists)f_2 p_2}{\exists} \rangle \text{cup} \text{sort}(P) \text{thinlist}(Q)$<br>$S = (f_1 p_1) \cup (f_2 p_2) \text{ — } \triangleq \text{thinlist, then } \triangleq \text{cup}$                                                                                                                      |                                                                                      |
| $\sqsubseteq$ <br>$\langle \frac{\text{F}(\exists)f_1 p_1}{\exists}, \frac{\text{F}(\exists)f_2 p_2}{\exists} \rangle (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \text{thinlist}(Q)$<br>$(\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \sqsubseteq \text{cup} \text{sort}(P) \text{ — (14.3b)}$                                                                |                                                                                      |
| $=$ <br>$\langle \frac{\text{F}(\exists)f_1 p_1}{\exists} \text{sort}(P), \frac{\text{F}(\exists)f_2 p_2}{\exists} \text{sort}(P) \rangle \text{merge}(P) \text{thinlist}(Q)$<br>$\langle X, Y \rangle (\text{sort}(P) \times \text{sort}(P)) = \langle X \text{sort}(P), Y \text{sort}(P) \rangle \text{ — bdm-prod-laws}$                                                         |                                                                                      |
| <br>$\sqsubseteq \text{F}(\text{sort}(P)) \text{listcp} \langle g_1, g_2 \rangle \text{merge}(P) \text{thinlist}(Q)$<br>$(14.3d) \text{ at } f_1, p_1 \text{ and at } f_2, p_2, \text{ then } X \langle g_1, g_2 \rangle \sqsubseteq \langle Xg_1, Xg_2 \rangle \text{ — bdm-prod-laws; } g_i \triangleq \text{list}(f_i) \text{filter}(p_i) \text{ — } \triangleq \text{thinlist}$ |                                                                                      |

## §14.4. The knapsack problem

(14.4a)

$\text{vol}, \text{wt} : \text{Item} \rightarrow \text{Real}, \text{value} \triangleq \text{list}(\text{vol}) \text{ sum}, \text{weight} \triangleq \text{list}(\text{wt}) \text{ sum}.$

$\text{subseq} = ([\text{nil}, \text{cons}] \cup [\text{nil}, \pi_2]) \text{ , , within } w \text{ the coreflexive on } xs \text{ with weight } xs \leq w, 0 \leq w.$

$R \triangleq \text{value} \geq \text{value}^\circ, Q \triangleq R \cap (\text{weight} \leq \text{weight}^\circ), P \triangleq R, .$

$\text{FA} = 1 + \text{Item} \times \text{A}, \text{listcp} = \text{wrap} + \text{cpr}, g_1 \triangleq \text{list}([\text{nil}, \text{cons}]) \text{ filter}(\text{within } w) = [\text{list}(\text{nil}), h_1],$   
 $g_2 \triangleq \text{list}([\text{nil}, \pi_2]) = [\text{list}(\text{nil}), h_2].$

$h_1 \triangleq \text{list}(\text{cons}) \text{ filter}(\text{within } w), h_2 \triangleq \text{list}(\pi_2).$

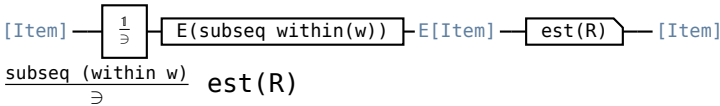
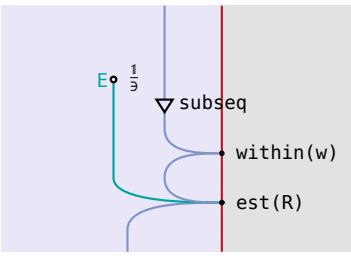
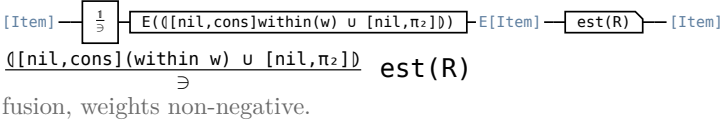
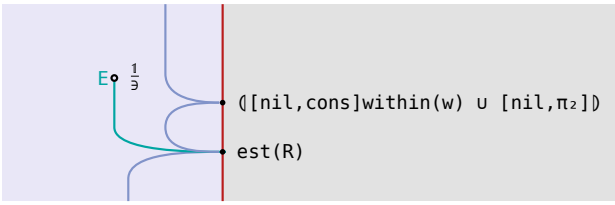
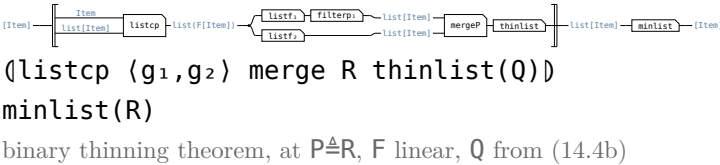
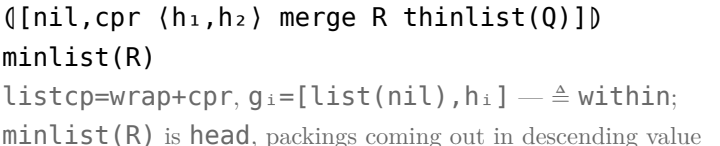
(14.4b)

| the law                                                                                                                                      | what it says                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| $(1 \times R) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) R$<br>FALSE                                      | not monotonic on R: a selection of greater value need not still fit once one more item goes in |
| $(1 \times Q) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) Q$<br>$(1 \times Q) \pi_2 \sqsubseteq \pi_2 Q$ , | both halves are monotonic on Q once ties in value are broken by weight                         |

$$\begin{aligned}
 & 0 \leq w \wedge (\forall i. 0 \leq \text{wt}(i)) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \\
 & \text{sortF}([\text{nil}, \text{cons}]R[\text{nil}, \text{cons}]^\circ) \text{listf}_1 \sqsubseteq P([\text{nil}, \text{cons}]) \text{sortP} \wedge \\
 & \text{sortF}([\text{nil}, \pi_2]R[\text{nil}, \pi_2]^\circ) \text{listf}_2 \sqsubseteq P([\text{nil}, \pi_2]) \text{sortP} \wedge \text{sortP } \text{filterp}_1 \sqsubseteq E(\text{within}(w)) \text{sortP} \wedge \\
 & F(\text{sortP}) \text{listcp} \sqsubseteq \text{cp}(F, [\text{Item}]) \text{sortF}(F(R)) \wedge (\text{sortP} \times \text{sortP}) \text{mergeP} \sqsubseteq \text{cup } \text{sortP} \wedge \text{sortP} \\
 & \text{thinlist} \sqsubseteq \text{thin}(Q(\text{vol}, \text{wt})) \text{sortP} \wedge \text{sortP } \text{minlist} \sqsubseteq \text{est}(R) \Rightarrow (\text{listcp}(\text{listf}_1 \text{ filterp}_1, \text{listf}_2) \\
 & \text{mergeP } \text{thinlist}) \text{minlist} \sqsubseteq \frac{\text{subseq } \text{within}(w)}{\supset} \text{est}(R)
 \end{aligned}$$

the knapsack problem, as a fold that thins the packings kept at each item

(14.4c)

| circuit                                                                             | Hinze–Marsden                                                                         |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|    |    |
|  |  |
|  |   |
|  |                                                                                       |

## §14.5. The paragraph problem

(14.5a)

```

Line=list+ Word, Para=list+ Line, FA=Word+Word×A, listcp=wrap+cpr.
new(a,xs)=[[a]]#xs, glue(a,xs)=[[a]#head(xs)]#tail(xs), partition≐([wrap wrap,new u glue]) : list+
Word→Para.

width≐([length,(length×1) plus succ]), 0≤length a, fits w the coreflexive on a line x with width x≤w, ok w the
coreflexive on [x]#xs with width x≤w.

white w x=w-width x, collect≐list(sqr) sum, waste w≐init list(white w) collect.

R≐(waste w)≤(waste w)°, Q≐Rn(head head°), P≐T, .

g1≐list([wrap wrap,new]), g2≐list([wrap wrap,glue]) filter(ok w), start≐wrap wrap wrap, h1≐list(new),
h2≐list(glue) filter(ok w).

```

(14.5b)

| the law                                                                                      | what it says                                                                                                             |
|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| (1×R) glue⊆glue R FALSE                                                                      | glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this |
| (1×Q) new⊆new Q<br>(1×Q) (glue (ok w))⊆glue (ok w)Q<br>cons monotonic on collect≤collect°. , | both halves are monotonic on Q once ties in waste are broken by the first line                                           |
| merge T=cat; P≐head prefix head° also serves                                                 | T needs no sorting at all, and prefix is a linear order on first lines of paragraphs of one input                        |

$$\begin{aligned}
& (\forall a. 0 \leq \text{len}(a)) \wedge (\forall a. \text{len}(a) \leq w) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{wrap} \\
& \text{wrap}, \text{new}] \tau [\text{wrap} \text{ wrap}, \text{new}]^\circ) \text{listf}_1 \sqsubseteq \text{P}([\text{wrap} \text{ wrap}, \text{new}]) \text{sortP} \wedge \text{sortF}([\text{wrap} \text{ wrap}, \text{glue}] \tau [\text{wrap} \\
& \text{wrap}, \text{glue}]^\circ) \text{listf}_2 \sqsubseteq \text{P}([\text{wrap} \text{ wrap}, \text{glue}]) \text{sortP} \wedge \text{sortP} \text{filterp}_2 \sqsubseteq \text{E}(\text{ok}(w)) \text{sortP} \wedge \\
& \text{F}(\text{sortP}) \text{listcp} \sqsubseteq \text{cp}(\text{F}, \text{list}^+((\text{list}^+ \text{Word}))) \text{sortF}(\text{F}(\tau)) \wedge (\text{sortP} \times \text{sortP}) \text{mergeP} \sqsubseteq \text{cup} \text{sortP} \wedge \text{sortP} \\
& \text{thinlist} \sqsubseteq \text{thin}(\text{Q}(\text{len}, w)) \text{sortP} \wedge \text{sortP} \text{minlist} \sqsubseteq \text{est}(\text{R}) \Rightarrow (\text{listcp}(\text{listf}_1, \text{listf}_2 \text{filterp}_2) \\
& \text{mergeP} \text{thinlist}) \text{minlist} \sqsubseteq \frac{\text{partition fits}(w)}{\supset} \text{est}(\text{R})
\end{aligned} \tag{14.5c}$$

a paragraph laid out as a fold that thins the layouts kept at each word

| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Hinze–Marsden |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| $ \begin{aligned} & \text{list}^+(\text{Word}) \xrightarrow{\frac{1}{3}} \text{E}(\text{list}^+(\text{Word})) \xrightarrow{\text{E}(\text{partition fits}(w))} \text{E}(\text{list}^+(\text{list}^+(\text{Word}))) \xrightarrow{\text{est}(\text{R})} \text{list}^+(\text{list}^+(\text{Word})) \\ & \frac{\text{partition list}^+(\text{fits } w)}{\supset} \text{est}(\text{R}) \end{aligned} $                                                                                                                                                                                                                                          |               |
| $ \begin{aligned} & \text{list}^+(\text{Word}) \xrightarrow{\frac{1}{3}} \text{E}(\text{list}^+(\text{Word})) \xrightarrow{\text{E}([\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}] \text{ok}(w))} \text{E}(\text{list}^+(\text{list}^+(\text{Word}))) \xrightarrow{\text{est}(\text{R})} \text{list}^+(\text{list}^+(\text{Word})) \\ & = \frac{([\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}] \text{ok}(w))}{\supset} \text{est}(\text{R}) \end{aligned} $ <p>fusion, every word fits on a line by itself.</p>                                                                                 |               |
| $ \begin{aligned} & \text{list}^+(\text{Word}) \xrightarrow{\frac{1}{3}} \text{E}(\text{list}^+(\text{Word})) \xrightarrow{\text{E}([\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}] \text{ok}(w))} \text{E}(\text{list}^+(\text{list}^+(\text{Word}))) \xrightarrow{\text{est}(\text{R})} \text{list}^+(\text{list}^+(\text{Word})) \\ & = \frac{([\text{wrap wrap}, \text{new}] \cup ([\text{wrap wrap}, \text{glue}] \text{ok } w))}{\supset} \text{est}(\text{R}) \end{aligned} $ <p>the algebra as <math>(f_1 p_1) \cup (f_2 p_2)</math>, <math>p_1 \triangleq 1 \text{ — } \triangleq \text{thinlist}</math>.</p> |               |
| $ \begin{aligned} & \text{listcp}(\text{listf}_1, \text{listf}_2) \text{cat thinlist}(\text{Q}) \\ & \sqsubseteq \text{minlist}(\text{R}) \end{aligned} $ <p>binary thinning theorem, at <math>\text{P} \triangleq \text{T}</math> with merge <math>\tau = \text{cat}</math>, <math>\text{Q}</math> from (14.5b)</p>                                                                                                                                                                                                                                                                                                                       |               |
| $ \begin{aligned} & ([\text{start}, \text{cpr}(\text{h}_1, \text{h}_2) \text{cat thinlist}(\text{Q})]) \\ & = \text{minlist}(\text{R}) \\ & \text{listcp} = \text{wrap} + \text{cpr}, g_i \text{ along the coproduct — } \triangleq \text{partition} \end{aligned} $                                                                                                                                                                                                                                                                                                                                                                       |               |

## §14.6. Bitonic tours

$\text{FA} = (\text{City} \times \text{City}) + (\text{City} \times \text{A})$ , the base functor of cons-lists of length at least two;  $\text{listcp} = \text{wrap} + \text{cpr}$ ;  $\text{tc} : \text{City} \times \text{City} \rightarrow \text{Real}$ , neither positive nor symmetric.

$\text{start } (a, b) = ([a, b], [a, b])$ ,  $\text{dropl } (a, ([b] \# xs, ys)) = ([a] \# xs, [a] \# ys)$ ,  $\text{dropr } (a, (xs, [b] \# ys)) = ([a] \# xs, [a] \# ys)$ ,  $\text{tour} \triangleq ([\text{start}, \text{dropl} \cup \text{dropr}])$ .

$\text{cost } (xs, ys) = \text{outcost } xs + \text{incost } ys$ ,  $\text{outcost } [a_0, \dots, a_n] = \text{tc } (a_0, a_1) + \dots + \text{tc } (a_{n-1}, a_n)$ ,  $\text{incost } [a_0, \dots, a_n] = \text{tc } (a_1, a_0) + \dots + \text{tc } (a_n, a_{n-1})$ .

$\text{next} \triangleq \text{tail head}$ ,  $\text{next2} \triangleq \text{next} \times \text{next}$ ,  $\text{head2} \triangleq \text{head} \times \text{head}$ ,  $\text{R} \triangleq \text{cost} \leq \text{cost}^\circ$ ,  $\text{Q} \triangleq \text{Rn}(\text{next2 next2}^\circ) \text{n}(\text{head2 head2}^\circ)$ ,  $\text{P} \triangleq \text{T}$ ,  $g_1 \triangleq \text{list}([\text{start}, \text{dropl}])$ ,  $g_2 \triangleq \text{list}([\text{start}, \text{dropr}])$ .

| the law                                                                                                                                                    | what it says                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| $(1 \times \text{R}) \text{dropl} \sqsubseteq \text{dropl } \text{R}$ FALSE<br>$(1 \times \text{R}) \text{dropr} \sqsubseteq \text{dropr } \text{R}$ FALSE | neither drop is monotonic on R: the two edges it adds and removes depend on head and next of both lists            |
| $(1 \times \text{Q}) \text{dropl} \sqsubseteq \text{dropl } \text{Q}$<br>$(1 \times \text{Q}) \text{dropr} \sqsubseteq \text{dropr } \text{Q}$             | both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input |

| $ \begin{aligned} & (\forall X. \forall Y. X \sqsubseteq Y \implies \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \\ & \text{sortF}([\text{start}, \text{droplFn}] \sqcup [\text{start}, \text{droplFn}]^\circ) g_1 \sqsubseteq P([\text{start}, \text{droplFn}]) \text{sortP} \wedge \\ & \text{sortF}([\text{start}, \text{droprFn}] \sqcup [\text{start}, \text{droprFn}]^\circ) g_2 \sqsubseteq P([\text{start}, \text{droprFn}]) \text{sortP} \wedge \\ & F(\text{sortP}) \text{listcp} \sqsubseteq \text{cp}(F, \text{dTour}(\text{City})) \text{sortF}(F(\tau)) \wedge (\text{sortP} \times \text{sortP}) \text{cat} \sqsubseteq \text{cup} \text{sortP} \wedge \text{sortP} \\ & \text{thinlist} \sqsubseteq \text{thin}(Q(\text{tc})) \text{sortP} \wedge \text{sortP} \text{minlist} \sqsubseteq \text{est}(R) \implies (\text{listcp}(g_1, g_2) \text{cat} \text{thinlist}) \text{minlist} \sqsubseteq \\ & \quad \underbrace{(\text{tourAlg})}_{\exists} \text{est}(R) \end{aligned} $ <p style="text-align: center;">a least-cost bitonic tour, as a fold that thins the tours kept at each city</p> |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Hinze–Marsden |
| $ \begin{aligned} & \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{\exists}} E(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E(\text{tourAlg})} E(\text{Journey}(\text{City})^2) \xrightarrow{\text{est}(R)} \text{Journey}(\text{City}) \\ & \frac{\text{tour}}{\exists} \text{est}(R) \end{aligned} $                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |
| $ \begin{aligned} & \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{\exists}} E(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E([\text{start}, \text{droplFn}] \sqcup [\text{start}, \text{droprFn}])} E(\text{Journey}(\text{City})^2) \xrightarrow{\text{est}(R)} \text{Journey}(\text{City}) \\ & = \frac{([\text{start}, \text{dropl} \sqcup \text{dropr}])}{\exists} \text{est}(R) \\ & \text{tour} \triangleq ([\text{start}, \text{dropl} \sqcup \text{dropr}]) \triangleq \text{tour} \end{aligned} $                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |               |
| $ \begin{aligned} & \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{\exists}} E(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E([\text{start}, \text{droplFn}] \sqcup [\text{start}, \text{droprFn}])} E(\text{Journey}(\text{City})^2) \xrightarrow{\text{est}(R)} \text{Journey}(\text{City}) \\ & \sqsubseteq (\text{listcp}(g_1, g_2) \text{cat} \text{thinlist}(Q)) \text{minlist}(R) \end{aligned} $ <p>binary thinning theorem, at <math>P \triangleq T</math> with merge <math>T = \text{cat}</math>, <math>Q</math> from (14.6b)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |               |
| $ \begin{aligned} & ([\text{start wrap}, \text{cpr}(\text{list}(\text{dropl}), \text{list}(\text{dropr})) \text{cat} \\ & \text{thinlist}(Q)] \text{minlist}(R) \\ & = \text{listcp} = \text{wrap} + \text{cpr}, g_i = [\text{list}(\text{start}), \text{list}(\text{drop}_i)] \triangleq \text{tour}; \\ & \text{quadratic, two tours added per step} \end{aligned} $                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |